Customer Behavior Analysis

Alfido Tech Internship – Task 1

Objective:

The objective of this project is to analyze customer purchasing behavior, identify customer segments, examine purchasing patterns, and detect churn risks. The insights gained from the analysis help improve customer engagement and support data-driven business decisions for Alfido Tech.

Dataset:

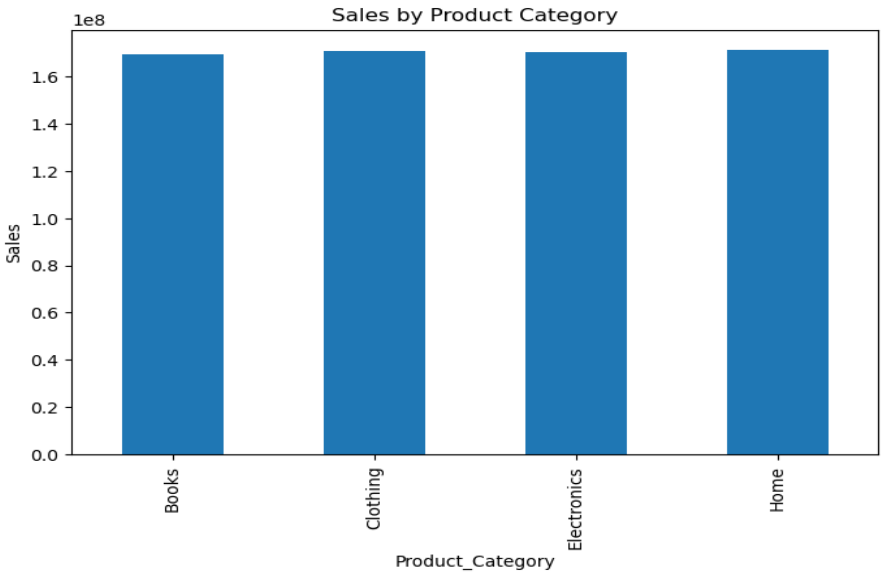
Dataset Source:Kaggle – Customer Behavior Analysis
Records: 250000
Columns:13

Data Cleaning:

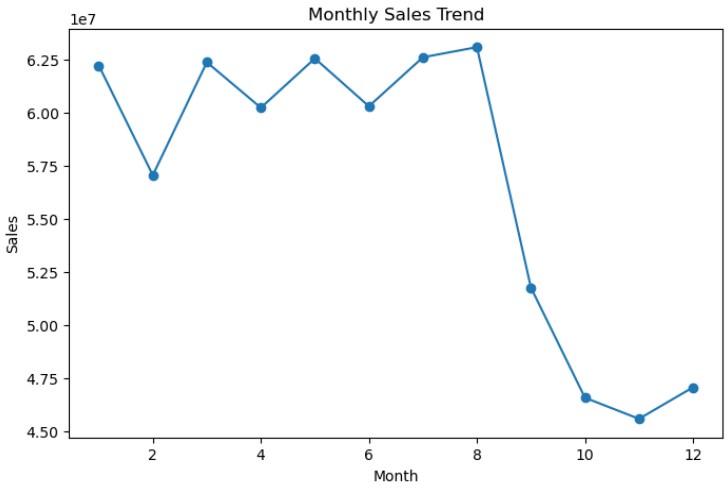
- Checked and handled missing values.
- Removed duplicate records.
- Renamed column names for consistency.
- Converted purchase date into datetime format.
- Created Month and Year features for analysis.

Visualizations:

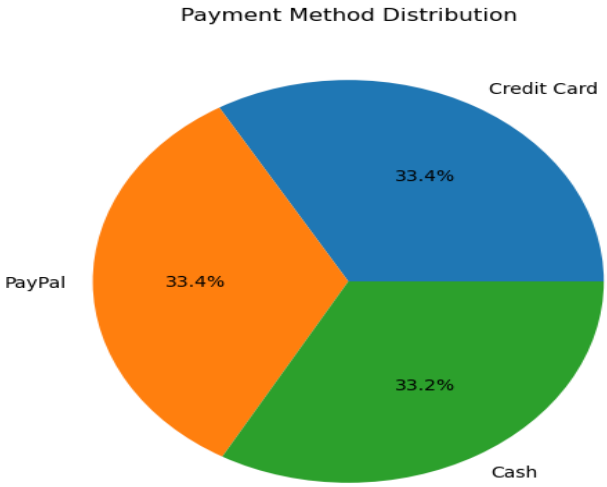
- Sales by Product Category



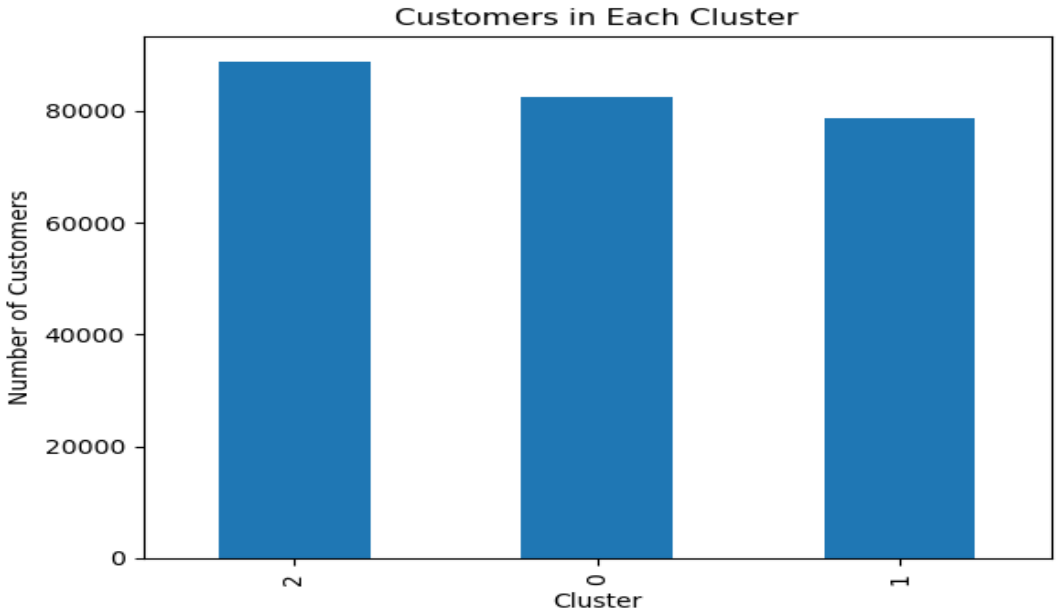
• Monthly Purchase Trend



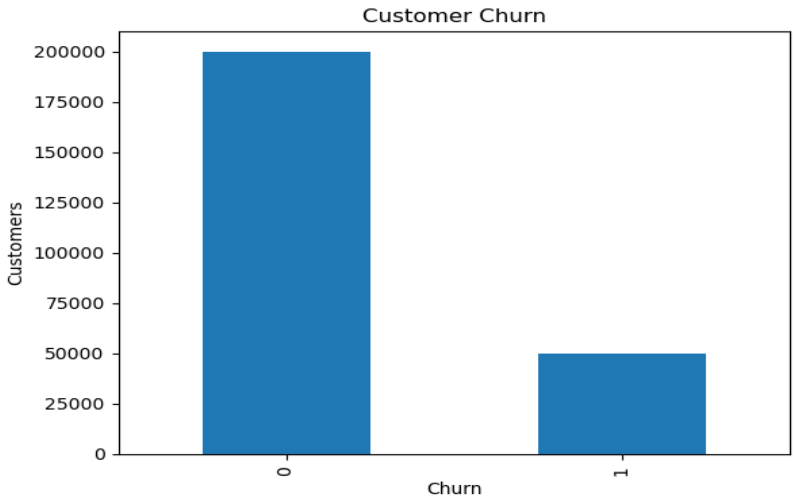
• Payment Method Distribution



• Customer Clusters



• Customer Churn Distribution



Key Findings:

• Customers were segmented into three distinct groups using K-Means clustering.

• Certain product categories generated higher purchase amounts.

• Monthly sales showed noticeable purchasing trends.

• Some customers are at risk of churn and require retention strategies.

• Digital payment methods are preferred by most customers.

Recommendations:

1. Introduce loyalty rewards for high-value customers.

2. Target churn-risk customers with personalized offers.

3. Promote best-selling product categories.

4. Use customer segmentation for personalized marketing.

5. Improve customer retention through targeted campaigns.

Conclusion:

Customer behavior analysis helped identify purchasing trends, customer segments, and churn risks. These insights can help Alfido Tech improve customer satisfaction, increase repeat purchases, and enhance business growth through data-driven decision-making.